School of Computer Science and Information Technology

Activity – 1

Activity : Infosys Springboard Certification

Subject : Programming in C

Date of submission : 6-10-2023

Submitted by:

Name: Bharath. K

Submitted to:

Dr. Pandikumar

JGi JAIN DEEMED-TO-BE UNIVERSITY | SCHOOL OF COMPUTER SCIENCE AND IT

Certificate

This is to certify that Ms. **BHARATH K** has satisfactorily completed activity 1 prescribed by JAIN (Deemed to be University) for the 1st semester degree course in the year 2023-2024.

Sr. No.	Criteria	Marks	Marks Obtained
1.	On-time Submission	10	
2.	Learning Outcomes	15	
3.	Report with details	15	
4.	Viva-Voce	10	
5.	Total	50	
6.	Conversation	15	

Signature of Student:____________

Date of Submission: 6-10-2023

Signature of Staff member: Prof. Pandikumar. S

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COURSE COMPLETION CERTIFICATE

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BHARATHK

for successfully completing the course

Programming in C

on October 5, 2023

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Senior Vice President and Head
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Issued on: Thursday, October 5, 2023
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Learning outcomes:

In the pursuit of expanding my knowledge and skill set in the field of computer programming, I embarked on a journey into the world of C programming. Over the course of 17 hours, I engaged in a comprehensive beginner-level C programming course designed to provide me with a solid foundation in this venerable programming language. Through a combination of theory, hands-on coding exercises, and practical applications, I aimed to acquire fundamental programming skills that would serve as the cornerstone for my future endeavors in software development.

In this report, I will outline the key learning outcomes and takeaways from this enlightening experience, highlighting the essential knowledge and competencies I have gained during the course.

1. Basic Understanding of C Programming:
After completing this course, I have gained a foundational understanding of the C programming language, including its syntax, data types, and basic program structure.

2. Programming Logic:
I have learned how to think logically and develop algorithmic solutions to simple problems using C. This includes using variables, operators, and control structures like loops and conditionals.

3. Variable Declaration and Data Types:
I can now declare variables of different data types such as integers, floating-point numbers, characters, and strings in C. I understand the importance of choosing the appropriate data type for different situations.

4. Input and Output:
I have acquired the skills to read input from the user and display output on the screen using functions like `printf` and `scanf`.

5. Functions and Modularity:
I can write and use functions to break down a program into smaller, manageable parts, promoting code reusability and maintainability.

6. Arrays:
I have learned how to declare, initialize, and manipulate arrays in C, allowing me to work with collections of data efficiently.

7. Control Structures:
I can use control structures like `if`, `else if`, `else`, `for`, `while`, and `do-while` loops to control the flow of a program and make decisions based on conditions.

8. Debugging Skills:
I have developed basic debugging skills to identify and fix common programming errors and issues in my C code.

9. Memory Management:
I understand the concept of memory allocation and deallocation in C, including dynamic memory allocation using functions like `malloc` and `free`.

10. File Handling:
I have learned how to read from and write to files in C, enabling me to work with external data sources and storage.

11. Coding Standards and Best Practices:
I am aware of the importance of adhering to coding standards and best practices to write clean, readable, and maintainable code.

12. Problem Solving:
I have enhanced my problem-solving skills by applying C programming concepts to real-world problems and projects.

13. Version Control:
I understand the basics of version control and can use tools like Git to manage and track changes in my code.

14. Documentation:
I can document my code effectively, including adding comments and writing clear, concise explanations of my program's functionality.

15. Collaboration:

I have gained the ability to collaborate with peers on programming projects, working together to achieve common goals.

In conclusion, the beginner-level C programming course has been a transformative learning experience that has equipped me with invaluable skills and knowledge in the world of programming. I have journeyed from a place of limited familiarity with C to now possessing a strong grasp of its fundamentals. The course has empowered me to think logically, solve problems systematically, and craft efficient code using C.

I have not only learned how to declare variables, work with control structures, and manage memory efficiently but also gained insights into best coding practices and the importance of documentation. Moreover, I have honed my debugging abilities and acquired a deeper appreciation for the intricacies of programming.

As I move forward in my programming journey, I am confident that the skills and insights gained from this course will continue to be an asset. Whether I am developing small-scale applications or tackling more complex programming challenges, the foundation I have built in C programming will remain a sturdy platform for my growth as a programmer.

In conclusion, I would like to express my gratitude to the instructors, mentors, and fellow learners who have made this learning experience enriching and rewarding. I look forward to applying these newfound skills and knowledge to future programming projects and challenges.

Report on Beginner-Level C Programming Course

Introduction:

The Beginner-Level C Programming course provided a comprehensive introduction to the fundamentals of the C programming language. Over the course of 17 hours, I embarked on a journey of learning and discovery, delving into the world of C programming to build a solid foundation in this venerable language.

Course Overview:

The primary objectives of the course were as follows:

1. Introduction to C: The course commenced with an introduction to the syntax, data types, and basic structure of the C programming language. It aimed to familiarize us with the essential elements of C.

2. Programming Logic: A significant focus of the course was on the development of logical thinking and problem-solving skills through the creation of algorithmic solutions in C. This module honed our ability to think systematically.

3. Core Concepts: We delved deep into core concepts such as variable declaration, data types, input/output mechanisms, and control structures. These building blocks formed the foundation of our C programming knowledge.

4. Functions and Modularity: To enhance code organization and reusability, the course introduced us to functions. We learned how to write, use, and manage functions effectively, promoting code modularity.

5. Arrays and Data Structures: As we progressed, we explored the concept of arrays and basic data structures in C. This knowledge proved essential for working with collections of data efficiently.

6. Memory Management: Memory allocation and deallocation were discussed in detail. We gained insights into dynamic memory allocation using functions like `malloc` and `free`, a critical skill for managing resources in C programs.

7. File Handling: To broaden our horizons, the course equipped us with the ability to read from and write to files. This skill allowed us to interact with external data sources and perform file operations in C.

Course Experience:

The course experience was both enriching and challenging. It offered a well-structured curriculum that ensured a gradual progression of difficulty, enabling learners to grasp complex concepts without feeling overwhelmed. Hands-on coding exercises were a pivotal part of the course, providing practical application of the theories learned.

The instructors were knowledgeable and supportive, providing clear explanations and guidance whenever needed. They fostered a collaborative learning environment, encouraging students to ask questions and engage in discussions.

One of the highlights of the course was the emphasis on coding standards and best practices. We were not only taught how to write functional code but also how to write clean, readable, and maintainable code. This aspect of the course was instrumental in developing a professional approach to programming.

Conclusion:

In conclusion, the Beginner-Level C Programming course has been a transformative learning experience. It provided a solid foundation in C programming, equipping me with essential skills and knowledge. I now have the ability to write clean and efficient code, solve problems systematically, and think logically.

The course not only covered technical aspects but also emphasized best coding practices, documentation, and collaboration. These skills are invaluable as I move forward in my programming journey.

I express my sincere gratitude to the instructors and fellow learners for their support and camaraderie throughout this course. With this newfound knowledge, I am excited about the possibilities and challenges that lie ahead in the field of software development. I look forward to applying these skills to create innovative solutions and contribute to the ever-evolving world of programming.